

Project Design Phase Problem - Solution Fit

Date	11 July 2026
Team ID	SWTID-2026-9508
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

The Problem-Solution Fit Canvas helps the team validate that the proposed Machine Learning solution genuinely addresses the real problems, constraints, and behaviour of the people it is designed for, before committing to build it.

1. CUSTOMER SEGMENT(S) Individuals applying for a credit card through a bank, and bank credit/loan officers reviewing those applications.	2. JOBS-TO-BE-DONE / PROBLEMS Applicant: get a fast, fair credit decision. Officer: assess applicant risk quickly and consistently across many applications.	3. TRIGGERS Needing a credit card for a purchase; officer facing a growing backlog of applications to review.	4. EMOTIONS: BEFORE / AFTER Before: anxious, uncertain (applicant); overwhelmed (officer). After: relieved, confident (applicant); in control (officer).
5. AVAILABLE SOLUTIONS Manual review by credit officers using internal checklists; generic rule-based approval thresholds. Both are slow and inconsistent.	6. CUSTOMER CONSTRAINTS Limited time to wait for a decision; officers have a high application volume and no standardized scoring tool.	7. BEHAVIOUR Applicant fills paper/online forms and waits for updates. Officer manually cross-checks income, debt, and credit history for each file.	8. CHANNELS of BEHAVIOUR 8.1 Online: bank website / application portal. 8.2 Offline: in-branch paper forms, phone follow-ups.
9. PROBLEM ROOT CAUSE Banks have no standardized, automated way to score creditworthiness. Every officer relies on personal judgement and informal rules, since no shared, data-driven decision tool exists - causing slow, inconsistent outcomes for both the bank and applicants.		10. YOUR SOLUTION A Machine Learning-based Credit Card Approval Prediction System: applicants' demographic and financial data is entered into a Flask web app, which runs a trained model (best of Logistic Regression, Decision Tree, Random Forest, XGBoost) to instantly return an Approved / Rejected decision - replacing slow, inconsistent manual review with a fast, consistent, data-driven process.	

Reference: Problem-Solution Fit Canvas methodology by Daria Nepriakhina / Amaltama.com
(<https://www.amaltama.com>)